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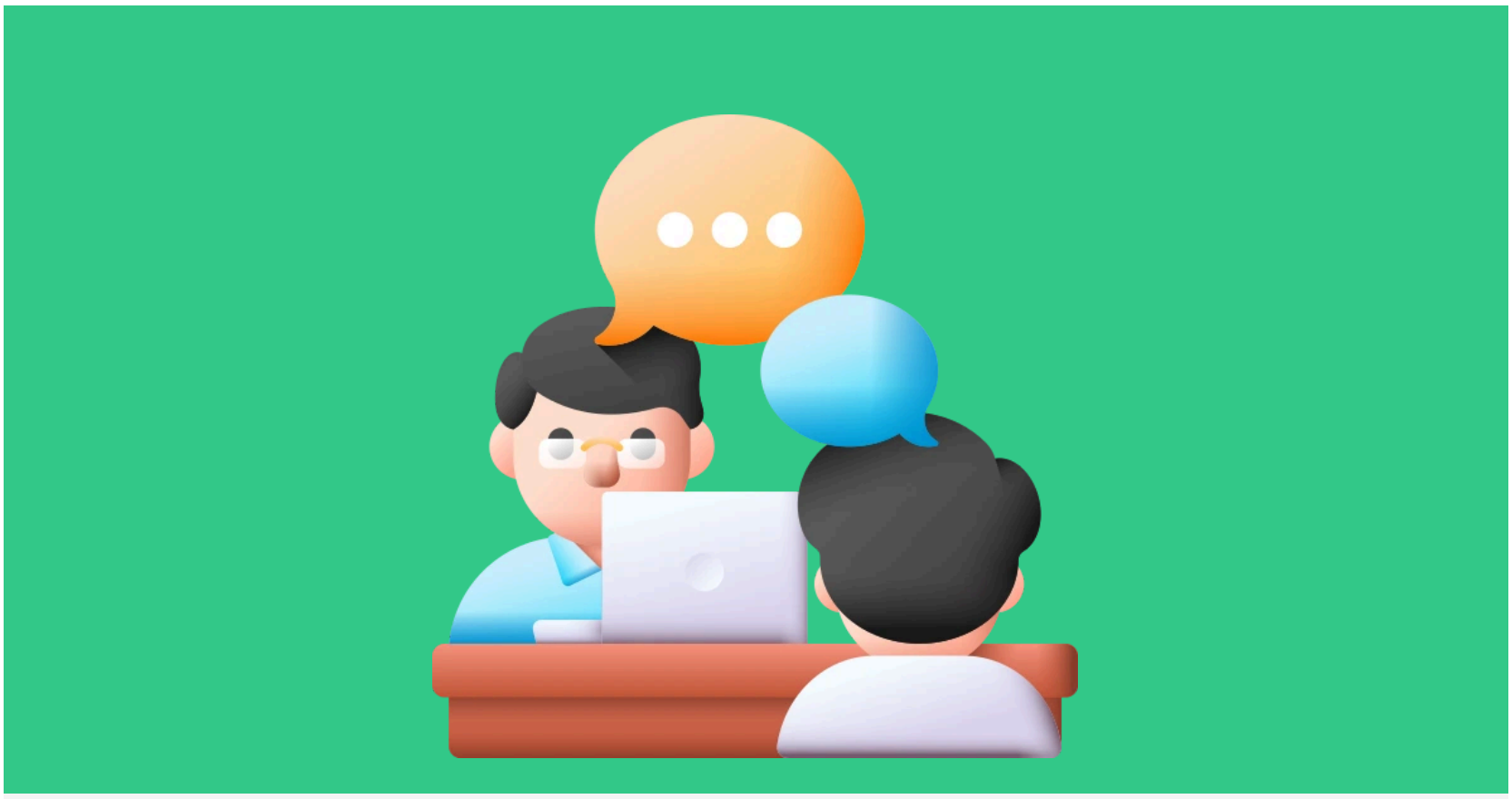
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TCS NQT Interview Questions and Answers



TCS NQT Interview Questions

- TCS NQT Technical Interview Questions
- TCS NQT HR Interview Questions

Also Checkout:

What is TCS NQT ?

TCS NQT Interview Questions

Important Topics for the TCS NQT Technical Interview 2026

1. **Programming Questions (C, Java, Python, C++):**

   You should know basic coding, syntax, logic-building, and sometimes write simple programs during the interview.

2. **OOPS Concepts:**

   Understanding the basics of Object Oriented Programming like inheritance, polymorphism, encapsulation, and abstraction is important.

3. **Computer Networks:**

   Basic questions about how computers connect and communicate, network layers, IP addressing, and protocols like TCP/IP.

4. **DBMS (Database Management Systems):**

   Expect questions about SQL queries, normalization, transactions, and database design basics.

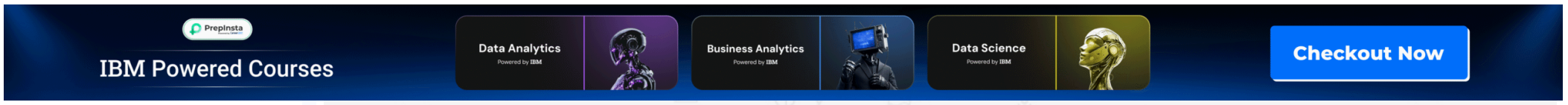
5. **DSA (Data Structures and Algorithms):**

   You must understand arrays, linked lists, stacks, queues, trees, sorting, and searching algorithms.

6. **Operating Systems:**

   Questions might include topics like memory management, scheduling algorithms, deadlocks, and basic concepts of Linux or Windows OS.

TCS has recently started asking basic questions about trending technologies like,



TCS NQT Coding Questions

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TCS NQT Interview Questions 2026

Sample TCS NQT Technical Interview Questions

Question 1. What is the difference between a process and a thread?

Answer:

- Process is an independent unit of execution with its own memory space, while a thread is a smaller unit within a process that shares the same memory and resources.
- Processes are isolated from each other and require more overhead to create and manage.
- Threads, being lightweight, are faster to create and switch but need careful synchronization due to shared memory. A crash in one process doesn't affect others, but a faulty thread can impact the whole process.

Question 2. Explain the concept of Object Oriented Programming (OOP).

Answer:

OOP is a way of designing and writing programs using objects.

An object is like a real world thing that has properties (called attributes) and can do things (called methods). These objects are created from classes, which are like blueprints or templates.

The main ideas in OOP are:

1. **Encapsulation:** Putting the data and actions that operate on the data together in one place (object). It also means protecting data from being changed directly.
2. **Abstraction:** Hiding complex details and only showing what's needed. **For example,** you don't need to know how a car works inside to drive it, just how to use the steering wheel.
3. **Inheritance:** A way to create new objects based on existing ones. It's like how a child inherits traits from their parents.
4. **Polymorphism:** It means using the same name for actions, but the action works differently depending on the object. **For example**, a dog and a cat both have a "speak" method, but they say different things.

Question 3. Differentiate between overloading and overriding.

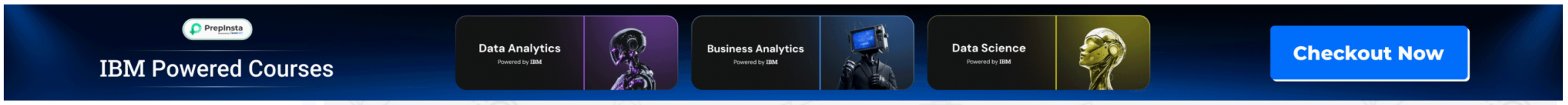
Answer:

- **Overloading:** Defining multiple methods with the same name but different parameters within the same class.
- **Overriding:** Redefining a method in a subclass that already exists in the parent class.

Question 4. What is a deadlock? How can it be prevented?

Answer:

A deadlock is a situation where two or more processes are waiting indefinitely for resources held by each other. Prevention strategies include resource ordering, avoiding circular wait, and using timeout mechanisms.



Normalization is the process of organizing data to reduce redundancy and improve data integrity. It involves dividing a database into two or more tables and defining relationships between them.

Question 6. What are ACID properties in DBMS?

Answer:

In **DBMS (Database Management System)**, **ACID** properties ensure reliable and consistent transactions. ACID stands for:

1. **Atomicity**: A transaction is all or nothing. Either all steps complete, or none do. If something fails, the database goes back to its original state.
2. **Consistency**: A transaction must take the database from one valid state to another, maintaining all rules (like constraints).
3. **Isolation**: Transactions happen independently. One transaction doesn't affect another, even if they run at the same time.
4. **Durability**: Once a transaction is complete, the changes are saved permanently, even if the system crashes.

Question 7. Explain the OSI model.

Answer:

The OSI Model is a 7 layer model that explains how data moves through a network.

Each layer has a specific job:

1. **Physical –** Sends raw bits (0s and 1s) over cables or wires.
2. **Data Link –** Handles direct connection between two devices (like error checking).
3. **Network –** Finds the best path to send data (IP addresses, routing).
4. **Transport –** Makes sure data is delivered fully and correctly (like TCP).
5. **Session –** Manages start, use, and end of communication sessions.
6. **Presentation –** Translates data format (like encryption or file formats).
7. **Application –** What users see (like web browsers, email apps).

It helps different systems talk to each other smoothly over a network.

Question 8. What is the difference between TCP and UDP?

Answer:

- **TCP (Transmission Control Protocol):** Connection-oriented, reliable, and ensures data delivery.
- **UDP (User Datagram Protocol):** Connectionless, faster, but does not guarantee delivery.

Question 9. What is a semaphore?

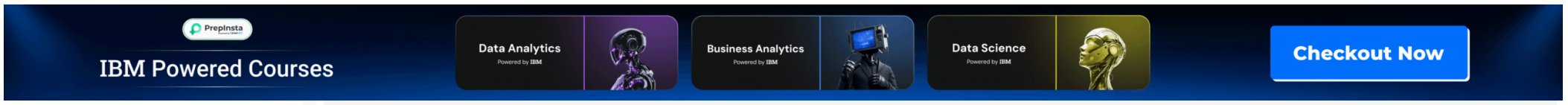
Answer:

A semaphore is a synchronization tool used to control access to a common resource in concurrent programming. It helps prevent race conditions.

Question 10. Explain the concept of virtual memory.

Answer:

Virtual memory is a memory management technique that gives an application the impression it has contiguous working memory, while in reality, it may be fragmented and spread across physical memory and disk storage.



Answer:

Paging is a memory management technique used in operating systems to manage how data is stored in RAM.

In simple words:

- **Paging** breaks the memory into small fixed size blocks:
  - **Pages** (in the process)
  - **Frames** (in physical memory/RAM)
- When a program runs, its pages are loaded into any available frames in RAM. The program doesn't need to be in one big block.

Question 12. Differentiate between stack and queue.

Answer:

Feature	Stack	Queue
Order	LIFO (Last In, First Out)	FIFO (First In, First Out)
Insertion	At the top	At the rear (end)
Deletion	From the top	From the front
Main Operations	`push()` to add, `pop()` to remove	`enqueue()` to add, `dequeue()` to remove
Access	Only one end (top) is accessible	Both ends are used (front and rear)
Real-life Example	Stack of plates, undo feature	People in a queue, printer task queue
Usage	Backtracking, function calls (recursion)	Scheduling, buffering, messaging

Question 13. What is Recursion?

Answer:

Recursion is a programming technique where a function calls itself to solve smaller parts of a problem until it reaches a base case (a stopping condition).

It's useful for problems that can be broken down into smaller, similar problems, like factorials, Fibonacci series, or tree traversals.

Example:

```
def factorial(n):
if n == 0: # base case
return 1
else:
return n * factorial(n - 1) # recursive call

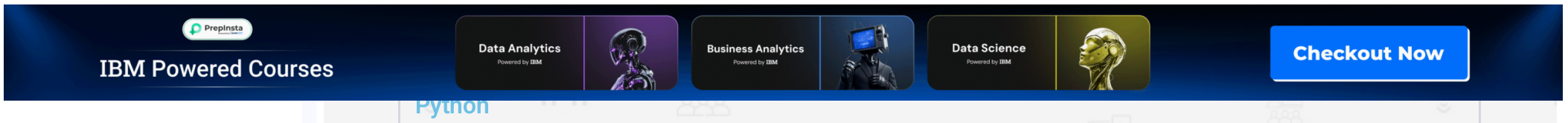
print(factorial(5))
# Output: 120
```

Question 14. Write a program to sort an array using bubble sort.

Answer:

C

C++




```
Python
```

Question 15. What is the difference between call by value and call by reference?

Answer:

- **Call by Value:** Copies the value of an argument into the formal parameter. Changes made inside the function do not affect the original value.
- **Call by Reference:** Copies the address of an argument into the formal parameter. Changes made inside the function affect the original value.

Question 16. Differentiate between malloc() and calloc().

Answer:

- **malloc():** Allocates a single block of memory of a specified size.
- **calloc():** Allocates multiple blocks of memory, each of the same size, and initializes all bytes to zero.

Question 18. What is the concept of normalization in databases.

Answer:

Normalization: The process of organizing data to reduce redundancy and improve data integrity.

It involves dividing a database into two or more tables and defining relationships between them.

Common normal forms include 1NF, 2NF, and 3NF.

Question 19. What is the concept of normalization in databases.

Answer:

Normalization: The process of organizing data to reduce redundancy and improve data integrity.

It involves dividing a database into two or more tables and defining relationships between them.

Common normal forms include 1NF, 2NF, and 3NF.

Question 20. What are the phases of the Software Development Life Cycle ?

Answer:

The SDLC consists of the following phases:

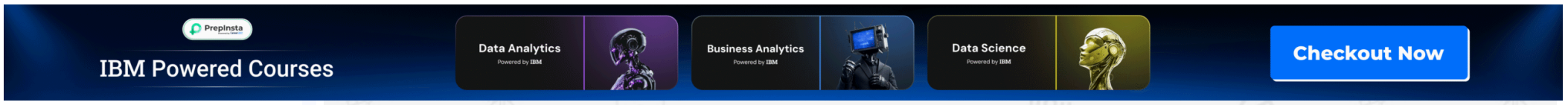
1. **Requirement Gathering** – Understanding client requirements.
2. **System Design** – Creating architecture and design specifications.
3. **Implementation (Coding)** – Writing actual code.
4. **Testing** – Verifying that the software works as intended.
5. **Deployment** – Releasing the product to users.
6. **Maintenance** – Fixing bugs and updating software post-deployment.

TCS NQT Interview Questions and Answers

Question 21. What is a foreign key in SQL?

Answer:

- A foreign key is a column or a set of columns in one table that establishes a link to the primary key of another table.



a Customers table.

This ensures that every order is linked to a valid customer.

Question 22. What is the difference between TCP and UDP in computer networks?

Answer:

1. **TCP, or Transmission Control Protocol,** is a connection oriented protocol that ensures reliable data delivery through error checking and acknowledgment mechanisms. It is used in applications like web browsing, email, and file transfers.
2. **UDP, or User Datagram Protocol,** is connectionless and does not guarantee delivery, but it is faster and more efficient for real time applications like video streaming and online gaming. The key difference lies in TCP's reliability versus UDP's speed.

Question 23. What is the difference between DELETE and TRUNCATE in SQL?

Answer:

1. DELETE is a SQL command used to remove specific rows from a table based on a condition, and the deleted data can often be rolled back if needed.
2. TRUNCATE is used to remove all rows from a table without logging individual row deletions, which makes it faster but irreversible in many cases.
3. While DELETE is useful for selective deletions, TRUNCATE is better suited for quickly clearing all records from a table.

Question 24. What is phishing in cybersecurity?

Answer:

1. **Phishing is a type of cyberattack** where attackers trick users into revealing sensitive information such as passwords, credit card numbers, or personal details by pretending to be a trustworthy entity, often through fake emails, messages, or websites.
2. The goal is usually to steal data or gain unauthorized access to systems.
3. Phishing can be avoided by not clicking on suspicious links, verifying the source of unexpected communications, and enabling multi factor authentication to add an extra layer of security.

Question 25. What are the main service models in cloud computing ?

Answer:

The main service models of cloud computing are Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS).

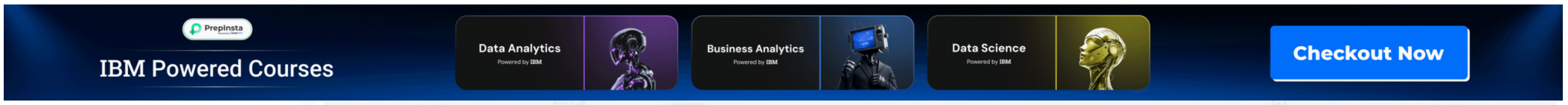
1. **IaaS** provides virtualized computing resources like virtual machines and storage.
2. **PaaS** offers a platform for developers to build and deploy applications without managing the underlying hardware.
3. **SaaS** delivers fully functional software applications over the internet, such as Gmail or Microsoft 365.

Question 26. What are the major deployment models in cloud computing ?

Answer:

Cloud computing offers several deployment models, each designed to meet specific business and security needs.

The four major deployment models are Public Cloud, Private Cloud, Hybrid Cloud, and Community Cloud.



customizable.

2. **Private Cloud** is dedicated to a single organization, offering greater control and security, often used by enterprises handling sensitive data.
3. **Hybrid Cloud** combines both public and private clouds, allowing data and applications to move between them for more flexibility and optimized performance.
4. **Community Cloud** is shared among organizations with common requirements, such as regulatory or compliance needs. Each model balances trade offs between cost, control, and security, and the choice depends on the organization's priorities and workload needs.

Question 27. How do you retrieve the second highest salary from a table in SQL?

Answer:

To find the second highest salary from a table, you can use either a subquery or use the LIMIT and OFFSET clause (in MySQL) depending on the database system.

One common way is using a subquery:

```
SELECT MAX(salary)
FROM employees
WHERE salary < (SELECT MAX(salary) FROM employees);
```

This query works by first finding the highest salary and then selecting the maximum salary that is less than that, effectively returning the second highest.

It's a frequently asked question because it tests understanding of subqueries and aggregate functions.

Question 28. What is the difference Between Compiler and Interpreter.

Answer:

- **Compiler** is a program that translates the entire source code of a programming language into machine code all at once, before the program runs. This results in a separate executable file that can be run independently of the source code. Because the entire code is compiled at once, execution is usually faster, but identifying errors can be harder, as they are all reported after compilation.
- **An interpreter,** on the other hand, translates and executes the source code line by line at runtime. It does not produce a separate executable file. This makes it slower in execution but easier to debug, as errors are caught and reported immediately during execution.

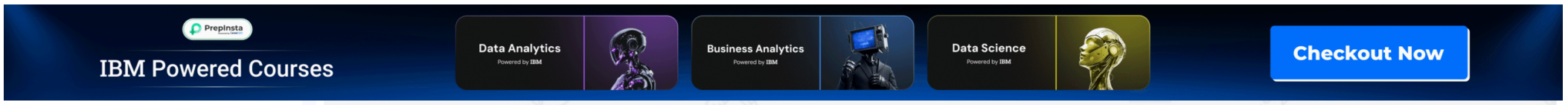
Example:

- Languages like C and C++ use compilers.
- Languages like Python and JavaScript use interpreters.

Question 29. Length of Longest Subarray with Sum Zero.

Answer:

C
C++
Java
Python



1. **Garbage collection in Java** is the process where the JVM automatically removes unused objects from memory to free up space.
2. When a Java program runs, it creates objects in a memory area called the heap.
3. If an object is no longer used or referenced, the garbage collector deletes it, so the memory can be used again.
4. This helps manage memory efficiently without the programmer having to do it manually.

TCS NQT HR Interview Questions

1. Tell me about yourself.

Answer:

I'm Aman Verma, a recent graduate in Engineering from Kalinga University. During my studies, I developed a strong interest in Data Analytics, which led me to work on projects like Project Management Tools and E-Commerce Platform.

I'm eager to apply my skills in a professional setting and contribute positively to your team.

2. Why do you want to join TCS?

Answer:

TCS is a global leader in IT services with a reputation for innovation and excellence. Joining TCS would provide me with the opportunity to work on diverse projects, learn from experienced professionals, and grow in a dynamic environment.

3. What are your strengths and weaknesses?

Answer:

My strength is my adaptability; I can quickly adjust to new environments and challenges. A weakness I've identified is my tendency to overcommit, but I'm learning to manage my time better to ensure quality work.

4. Are you willing to relocate?

Answer:

Yes, I'm open to relocation. I believe that working in different locations can provide valuable experiences and learning opportunities.

5. Can you work night shifts?

Answer:

Yes, I'm flexible with work timings and understand that certain projects may require night shifts. I'm prepared to adjust my schedule as needed.

6. Why should we hire you?

Answer:

I bring enthusiasm, a strong work ethic, and a willingness to learn. I'm committed to contributing positively to your team and growing with the company.

7. Where do you see yourself in 5 years?

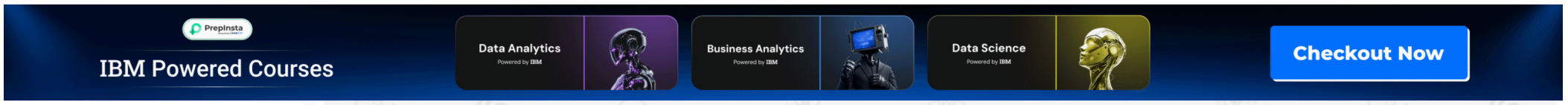
Answer:

In five years, I aim to have grown into a role with greater responsibility, contributing significantly to projects and mentoring new team members.

8. Do you prefer working in a team or individually?

Answer:

I enjoy both. Working in a team allows for collaboration and diverse ideas, while individual tasks help me focus and take ownership.



During my final year project, I came accross a technical issue that threatened my project's deadline. I researched solutions, consulted with mentors, and coordinated with my team to implement a fix, ensuring timely completion.

At that time I realized the panicking is not an option for us, we should always try to be patient and try to resolve any issue calmly.

10. What do you know about TCS?

Answer:

TCS, or Tata Consultancy Services, is a leading IT services company headquartered in Mumbai. It offers consulting, software development, and business solutions globally.

TCS NQT Interview Questions

11. What motivates you?

Answer:

The opportunity to learn new things and solve challenging problems motivates me. Achieving goals and contributing to a team's success gives me a sense of fulfillment and leadership.

12. How do you handle stress and pressure?

Answer:

I stay calm and focused, breaking down tasks into manageable parts and prioritizing effectively. I also ensure to take short breaks to maintain productivity and clarity.

13. What are your hobbies?

Answer:

I enjoy reading tech blogs to stay updated, traveling and I also like playing chess, which helps me improve my strategic thinking and concentration.

14. Do you have any backlogs?

Answer:

No, I have cleared all my semester's exam successfully without any backlogs.

15. What is your greatest achievement?

Answer:

Leading my college's tech fest as the coordinator, where I managed a team and successfully organized multiple events, enhancing my leadership and organizational skills. Without any problem I managed all the events.

16. How do you handle failure or setbacks?

Answer:

I see failure as an opportunity to learn. During a college project, my code didn't run during the final demo. I took feedback from my mentor, restructured the logic, and the second attempt worked perfectly. That experience taught me to double check and test under different scenarios.

17. If you face a conflict with a teammate, how would you handle it?

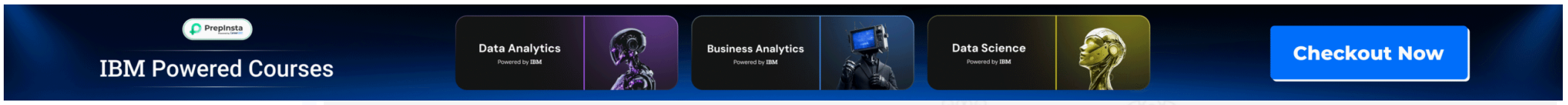
Answer:

I would first try to understand their perspective and keep the conversation respectful. I believe communication can solve most conflicts. If it doesn't help, I'd involve a neutral senior or mentor to mediate.

18. How would you describe your communication skills?

Answer:

I believe I have clear and polite communication skills. I make sure to listen actively and explain my thoughts in a simple and structured way. I've practiced this during group projects, presentations, and coding contests.



I stay calm and break tasks into smaller chunks with mini deadlines. I also avoid multitasking and focus on one thing at a time. If I feel stuck, I take a short break or talk to teammates for support, it helps me regain focus.

20. Do you have any questions for us?

Answer:

Yes, I'd love to know more about the learning and upskilling programs for freshers at TCS.
or
Can you share more about the kind of projects freshers are typically assigned to?

FAQ's

- 1. What is TCS NQT?
- 2. Who is eligible to apply for TCS NQT 2026?
- 3. What is the exam pattern for TCS NQT 2026?
- 4. Is there any negative marking in the TCS NQT exam?
- 5. How can I apply for the TCS NQT exam?

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